

AI-GENERATED MODEL PAPER

Subject: Database Management Systems

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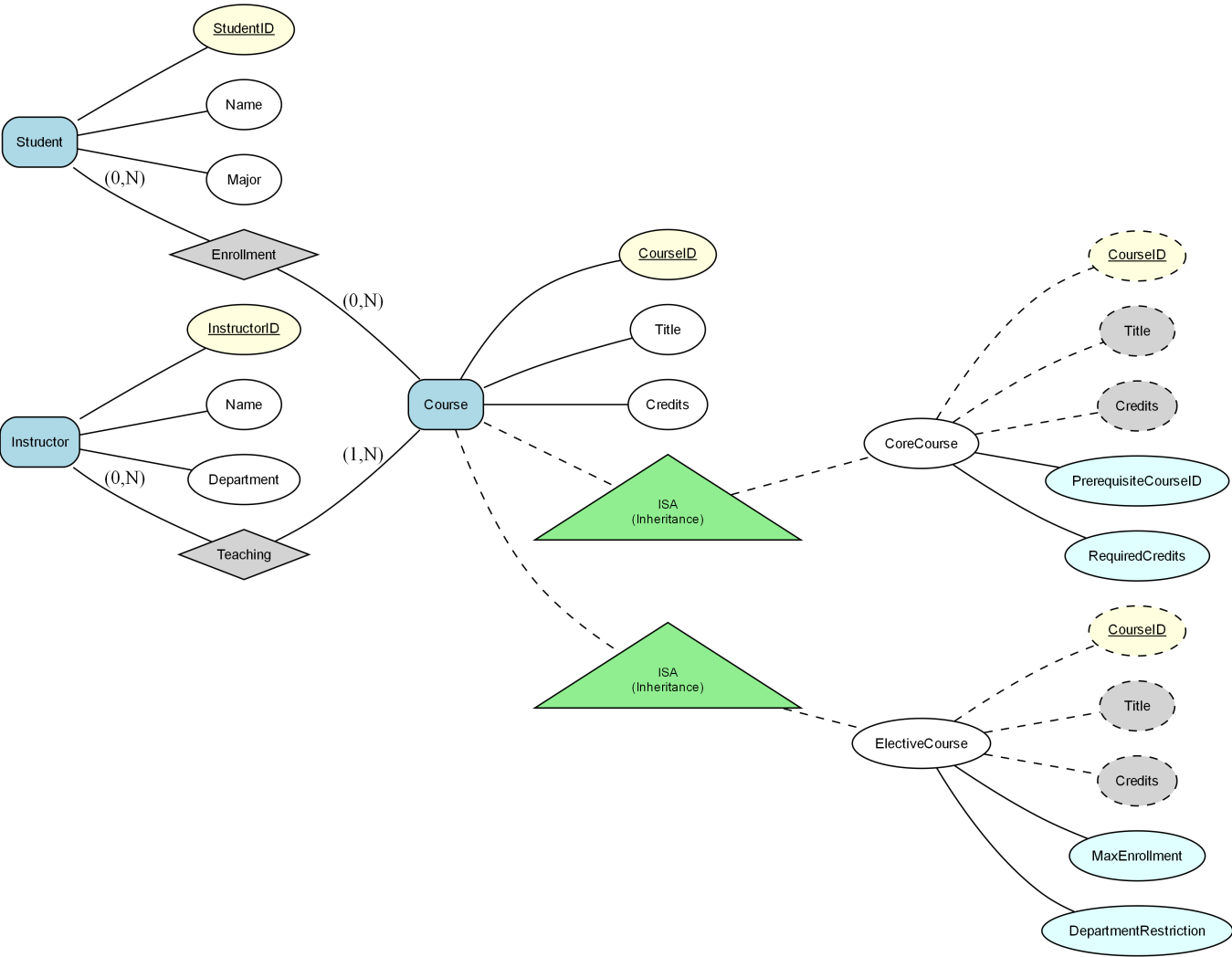
Total Marks: 100

Examination Paper

Question Q1 (20 marks)

Consider the following Enhanced Entity-Relationship (EER) diagram:

Figure: EER



- a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17)
- b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3)

Question Q2 (15 marks)

Consider a relation $R(\text{StudentID}, \text{CourseID}, \text{Instructor}, \text{Semester})$ with the following set of functional

dependencies F over R: $F = \{\text{StudentID} \rightarrow \text{CourseID}, \text{CourseID} \rightarrow \text{Instructor}, \text{Semester} \rightarrow \text{CourseID}\}$.

- a) Draw the functional dependency diagram. (1)
- b) Use the attribute closure and identify the primary key for the given relation. (1)
- c) If we insert a row $\langle 101, \text{CSE101}, \text{Dr. Smith}, \text{Fall} \rangle$ into the given relation, what is the update anomaly that is violated? Give reasons. (2)
- d) Assume that the relation has a row, $\langle 102, \text{CSE102}, \text{Dr. Jones}, \text{Spring} \rangle$. If we remove the course CSE102, what is the possible update anomaly that can be violated? Give reasons. (2)
- e) Using the above functional dependencies, design a set of 3NF relations for the above given relation. Show clearly each stage in deriving the 3NF relations. Also identify the normal form at each of these stages. (3)
- f) Do the update anomalies of part (c) and (d) hold in the 3NF relations? Discuss briefly. (2)
- g) Are the 3NF relations derived in part (e) still in 3NF? Justify your answer. (2)
- h) Are the 3NF relations derived in part (e) in BCNF? If not, bring the relations to BCNF. (2)

Question Q3 (25 marks)

A university is developing a database system to manage its courses and student enrollments. Each course has a unique ID and a title, while each student has a unique student ID and associated personal information. The system is intended to facilitate enrollment tracking and course management. The system is built upon a relational database model, where SQL Data Definition Language (DDL) is employed to define the structure of the database. The administration of the database involves assigning roles to various personnel including administrators, course coordinators, and data entry operators.

- a) In a relational database, each table is defined by a schema that includes its fields and data types. Accept or refute the above statement justifying your answer. (2)
- b) There are several different statements in SQL DDL to enforce data integrity. Which type of statements are used in the code segment given below? Briefly explain when this type of statement will be used. (2)
- c) Briefly explain the options available for transferring data between different tables within the same database. (2)
- d) Briefly explain how user privileges and roles can be managed in a MySQL database. (3)
- e) A financial institution is developing a robust database system to manage its clients'

sensitive financial data. Different roles are assigned to various team members for database administration and management. Sarah is the senior database administrator tasked with overseeing the entire database system's creation and maintenance. Tom and Emily are junior database administrators under Sarah's supervision. Tom manages the client accounts database, while Emily handles the transactions database. Nathan is a database developer reporting to Tom and Emily. His responsibilities include designing and implementing database schemas for client accounts and transactions. Grace and Michael are data entry operators responsible for inputting client information and transaction records into the system. Write a T-SQL statement to create a login for Sarah with windows authentication. (3)

f) Provide Jessica with the necessary permissions to execute all administrative commands within the database using a predefined role. (2)

g) Assuming Lisa's username is 'lisa.l', write SQL statements to assign her the responsibility of managing the student database using a user-defined role. (5)

h) Assuming Oliver's username is 'oliver.o', write SQL statements to grant him permission to create tables and alter existing ones. (3)

i) Assuming Maria's username is 'maria.m', write SQL statements to allow her to perform data entry tasks on both the courses and student databases. (3)

Question Q4 (40 marks)

A library system manages books, members, and loans. Each member can borrow multiple books, and each book can be borrowed by multiple members over time. The system tracks the date of loan and return. There are specific genres for books, and each member is assigned a unique membership ID alongside their contact information. Your task is to analyze this scenario and perform the required actions below.

a) Identify the entities in the scenario and provide an attribute list for each entity. (10)

b) Clearly define the relationships between the identified entities, indicating cardinalities. (10)

c) Draw an ER diagram representing the entities, relationships, and their attributes. Ensure to include appropriate cardinalities and relationship names. (10)

d) Map the ER model to a relational schema, specifying primary and foreign keys in your design. (10)